Dense Caption Reconstruction: Video In-filling with Language Models

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Subtitle
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Abstract

Recent Video-LLMs typically treat video understanding as a continuous stream of visual encoding. However, real-world events often follow structured, semantic scripts that pre-trained language models can predict without immediate visual evidence. In this work, we investigate the boundary between **semantic inference** (what must happen) and **visual perception** (what actually happened) through a novel *Caption Reconstruction* comparative framework. We use two independent methods—a text-based LLM and a Visual-Interpolation baseline method—to reconstruct missing segments from masked videos. Our analysis showcases a spectrum of narrative predictability across video domains: while stochastic events (e.g., nature) require visual grounding, procedural events (e.g., farming, manufacturing) allow text-only models to in-fill accurate reconstructions. This framework serves as a diagnostic tool for measuring multimodal information density and suggests potential for temporal semantic compression in next-generation video architectures.

1. INTRODUCTION

The promise of multimodal AI is the fusion of visual perception with semantic reasoning. Yet, in current Video-LLM architectures, this fusion is often brute-forced: models ingest massive sequences of visual tokens regardless of the information content. This approach ignores a fundamental property of the physical world: narrative predictability.

Consider a video of a person chopping an onion. If the next scene is missing, a human (or an LLM) can predict with high confidence that the onion was fried, the peppers were added, and the dish was stirred. The exact visual details are *stochastic residuals*, but the *semantic action* is procedurally deterministic. We propose a **Comparative Reconstruction Framework** to quantify this "Semantic Gap" between text-based inference and visual-only interpolation.

2. METHODOLOGY

2.1. Problem Formulation

We represent a video V as a sequence of T semantic units, where each unit u_t consists of a visual embedding v_t and a textual caption c_t. The task is to mask a subset of these units $M \subset \{1 \dots T\}$ and reconstruct the missing semantic content. Our target is the **semantic embedding** of the missing segment.

2.2. Comparative Pathways

Text Logic: Tests the limit of **Semantic Predictability**. It receives observed segments $C_{\text{obs}} = \{(t, c_t) \mid t \notin M\}$ and employs an LLM to predict the text $\hat{c}_t$, which is then encoded: $\hat{e}_{\text{text}} = \text{Encoder}(\hat{c}_t)$.

Visual Continuity: Tests **Visual Fidelity**. It receives visual embeddings $V_{\text{obs}} = \{v_t \mid t \notin M\}$ and reconstructs $\hat{e}_{\text{vis}}$ via non-parametric weighted averaging of neighboring vectors.

2.3. Evaluation: Predictability Delta

To normalize across different embedding densities, we define the **Predictability Delta (Δ)**:

$$\Delta = \text{Rank}(\hat{e}_{\text{text}}) - \text{Rank}(\hat{e}_{\text{vis}}) \tag{1}$$

where $\Delta \ll 0$ indicates **Semantic Dominance** (predictable logic) and $\Delta \gg 0$ indicates **Visual Necessity** (stochastic change).

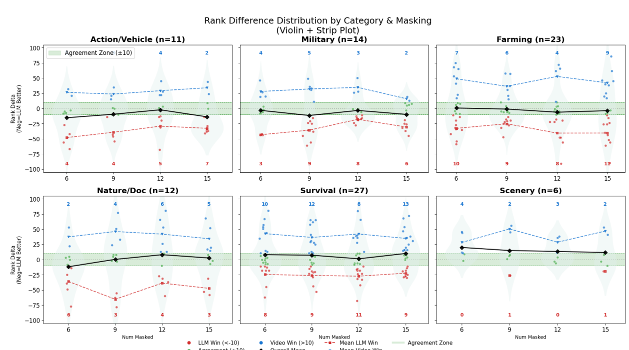

Figure 1: The Predictability Spectrum. Distributions of Δ across video categories show clear shifts between Procedural and Stochastic domains.

3. EXPERIMENTS

We evaluate on a dataset derived from WildQA, spanning 100 diverse videos.

3.1. Domain Analysis

Procedural Resilience: Categories like *Farming* and *Military* exhibit instances of extreme semantic dominance ($\Delta < -50$). The LLM successfully reconstructs intermediate steps even when the visual jump is large.

Stochastic Dependence: *Nature* and *Scenery* show a heavy tail toward visual necessity ($\Delta > 50$), as low-level physical changes are hard to describe textually.

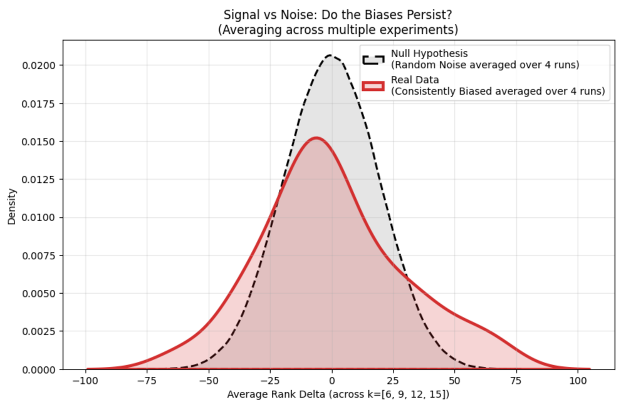

Figure 2: Robustness Check. Comparison of Average Rank Delta across masking levels ($k = 6, 9, 12, 15$). The distribution is significantly wider than random chance, indicating persistent modality advantages.

4. DISCUSSION

Our findings support the view that VLMs suffer from information redundancy. For procedural content, the visual stream is often semantically redundant. Future work will investigate the "Event Horizon"—the point at which even procedural scripts become chaotic and both models fail.

References

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